ADITYA UNIVERSITY

Worksheet - 02

Course: Aptitude **Topic**: LCM & HCF

1. Two numbers have an LCM of 252 and an HCF of 7. If one of the numbers is 28, find the other number.
   A) 49 B) 56 **C) 63** D) 84
   Sol: Express numbers as 7a and 7b (with a and b co-prime). Since 28 = 7×4, set a = 4; using LCM = 7ab = 252 gives b = 9. The other number is 7×9 = 63.

2. LCM of two numbers is 91 and HCF is 1. If one of the numbers is 13. Find the other number?.
   A) 13 **B) 7** C) 637 D) 1/13
   Sol:
   Given that L.C.M = 91, HCF=1 and one of the number a=13, b=?
   We know that the relation between LCM, HCF as LCM x HCF = a x b
   So 91 x 1 = 13 b => b=(91x1)/13 => 7

3. If two numbers have HCF = 6 and LCM = 72, what is the product of the two numbers?
   A) 432 B) 468 C) 4320 D) 43200
   Sol: The product equals HCF × LCM, so 6 × 72 = 432.

4. Find the least number which when divided by 15,27,35,42 leaves in each case a remainder 9?.
   A) 1881 **B) 1899** C) 954 D) 963
   Sol:

   Required number = LCM of 15, 27, 35, 42 + remainder

   = 1890 + 9 =1899

   (or) By using options => 1899 – 9 = 1890.
   Check the divisibility rules of 9 & 7 we discussed in the class.

5. Given that the product of two numbers is 1050 and their HCF is 7, what is their LCM?
   A) 147 **B) 150** C) 210 D) 1050
   Sol: Using the formula: Product = HCF × LCM, we have LCM = 1050 ÷ 7 = 150.

6. What is the LCM of 45 and 60?
   A) 90 B) 120 **C) 180** D) 360
   Sol: 45 = 3^2×5 and 60 = 2^2×3×5 yield LCM = 2^2×3^2×5 = 180.

7. Two numbers have an HCF of 7, an LCM of 84, and their sum is 91. What are the two numbers?
   A) 7 and 84 B) 14 and 77 C) 21 and 70 D) 28 and 63
   Sol: Represent the numbers as 7a and 7b with a + b = 91 ÷ 7 = 13 and product a×b = (7a×7b)/7 = 84 ÷ 7 = 12. The only positive integers summing to 13 and multiplying to 12 are 1 and 12. Thus, the numbers are 7×1 = 7 and 7×12 = 84.

8. Find the LCM of 8, 9, and 12.
   A) 72 B) 96 C) 108 D) 144
   Sol: 8 = 2^3, 9 = 3^2, and 12 = 2^2×3. The LCM is 2^3×3^2 = 72.

9. Determine the HCF of 65, 130, and 195.
   A) 5 B) 13 **C) 65** D) 195
   Sol: 65 = 5×13, 130 = 2×5×13, and 195 = 3×5×13; all share 5 and 13, so the HCF is 65.

10. Find the smallest number which when decreased by 5, is exactly divisible by 20, 28, 35, 105.
    A) 420 **B) 425** C) 415 D) 405
    Sol:
    Smallest number = LCM of 20, 28, 35, 105 + decreased number
    = 420 + 5 => 425

11. Find the greatest number that will divide 40, 48, and 72 leaving remainders 4, 6, and 8, respectively.
    A) 8 B) 10 **C) 2** D) 16
    Sol: Subtract remainders from each number and find HCF of (40−4), (48−6), (72−8), which is HCF(36, 42, 64) = 2.

12. Find the largest number that exactly divides 120, 150, and 180 leaving remainders 3, 5, and 7 respectively.
    A) 1 B) 7 C) 9 D) 11

Sol: Subtract remainders from each number and find HCF of (120−3), (150−5), (180−7), which is HCF(117,145,173).

13. Two numbers have an HCF of 12 and an LCM of 360. If their sum is 132, what are the numbers?
A) 60 and 72 B) 48 and 84 C) 36 and 96 D) 54 and 78
Sol: Express the numbers as 12·m and 12·n with m+n = 132/12 = 11 and m·n = (360×12)/(12^2) = 30. The only positive integers satisfying m+n = 11 and m·n = 30 are 5 and 6, so the numbers are 60 and 72.

14. Three numbers have an HCF of 4 and an LCM of 240. If two of the numbers are 20 and 24, what is the third number?
A) 40 B) 60 **C) 80** D) 100
Sol: The LCM of 20 and 24 is 120. To "lift" the overall LCM to 240, the third number must contribute an extra factor (here, an extra 2). Testing shows 80 works since 80 = 2^4·5 and the overall HCF remains 4.

15. Find the largest number that divides 117, 141, and 183 leaving the same remainder in each case.
A) 3 **B) 6** C) 9 D) 12
Sol: Subtracting the numbers pairwise yields differences 24, 42, and 66. Their HCF is 6; hence, the required divisor is 6.

16. In a race the finishing times (in minutes) of two runners are integers. Their LCM is 180 and their HCF is 6. If the faster runner's time is 30 minutes, what is the slower runner's time?
A) 36 B) 40 C) 42 D) 48
Sol: Using the relation (a × b) = HCF × LCM, we have 30 × (other) = 6 × 180 = 1080, so the other time is 1080/30 = 36 minutes.

17. The least square number which is exactly divisible by 5,6,8, 9 and 10 is
A) 1600 B) 2500 **C) 3600** D) 4900
Sol:

) By using prime factorisation Method.

$5 = 5^1$, $6 = 2^1 \times 3^1$, $8 = 2^3$, $9 = 3^2$, $10 = 2^1 \times 5^1$

L.C.M = $5^1 \times 2^3 \times 3^2$

for getting perfect square number we can change the powers of each base into even value by multiplying with same bases

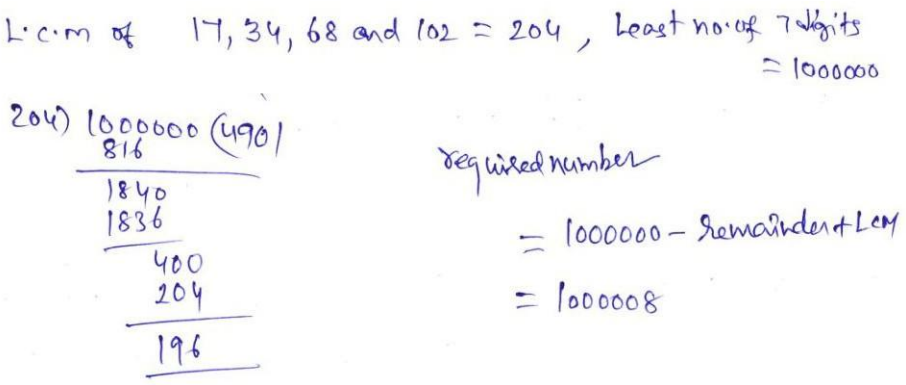
$5^1 \times 2^3 \times 3^2 \Rightarrow 5^1 \times 2^3 \times 3^2 \times 5^1 \times 2^1$

∴ required number = 3600

18. Find the least number of 7 digits which is exactly divisible by 1 7, 34, 68 and 102

A) 1000212 B) 1000204 C) 1000002 **D) 1000008**

Sol:



19. Find the LCM and HCF of 7/9, 63/8, 9/7.

A) 63, 1/504 B) 504, 1/63 C) 9, 1/72 D) 7, 1/56

Sol:1. H.C.F. =H.C.F. of Numerators / L.C.M. of Denominators

2. L.C.M. =L.C.M. of Numerators / H.C.F. of Denominators

LCM=(LCM of numerator) / (HCF of denominator)

LCM of (7, 63, 9) / HCF of (9, 8, 7)=63

HCF of numerators=1

LCM of denominators=504

Thus HCF= 1/504.

Hence, the answer is 63, 1/504.

20. Find the LCM and HCF of 6/1, 60/17, 12/17.

A) 60, 6/17 B) 120, 6/34 C) 60, 12/17 D) 30, 6/17

Sol: For the LCM and HCF of fractions, there is a formula.

L.C.M.= L.C.M. of numerators/ HCF of denominators

H.C.F.= H.C.F. of numerators/ LCM of denominators

L.C.M. = L.C.M.(6, 60, 12)/ HCF(1,17,17)

= 60/1= 60

H.C.F.= H.C.F.(6,60,12)/ L.C.M.(1,17,17)

= 6/17

21. Find the smallest number of five digits exactly divisible by 16, 24, 36 and 54.(Wipro)

A) 10368 B) 10080 C) 10800 D) 10224

Sol: Smallest number of five digits is 10000.

Required number must be divisible by L.C.M. of 16, 24, 36, 54, i.e., 432.

On dividing 10000 by 432, we get 64 as remainder.

∴ Required number = 10000 + (432 – 64) = 10368.

22. The traffic lights at three different road crossings change after every 48 sec., 72 sec. and 108 sec. respectively. If they all change simultaneously at 8:20:00 hours, then at what time will they again change simultaneously? **(Infosys)**

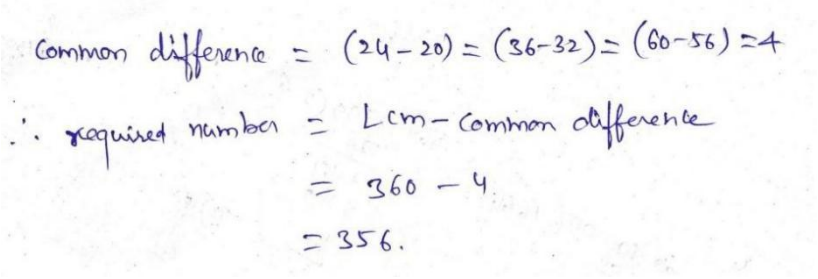
A) 7:50:44 hrs B)8:30:35 hrs C)8:15:12 hrs **D)8 : 27 : 12 hrs.**

Sol: Interval of change = (L.C.M. of 48, 72, 108) sec. = 432 sec.
So, the lights will again change simultaneously after every 432 seconds, i.e., 7 min. 12 sec. Hence, next simultaneous change will take place at 8 : 27 : 12 hrs.

23. Find the smallest number which when divided by 24, 36 and 60 leaves 20, 32 and 56 as remainders respectively. **(Infosys)**

A) 356 B)360 C)364 D)372

Sol:



24. Two Choco sticks are 48 cm and 64 cm long. If we have to make them into a number of Choco sticks of equal size, then minimum number of similar Choco sticks is _____ **(Wipro)**

A) 15 B)10 C)12 **D)7**

Sol: Length of the two Choco sticks are 48 cm and 64 cm.
We have to make minimum number of similar Choco sticks which means that each Choco stick should have maximum length.
Therefore we can find maximum length by taking HCF(48,64) = 16.
Each Choco stick should have length 16 cm.
Therefore number of Choco sticks = (48+64)/16 = 112/16 = 7 Choco sticks
Hence, the answer is 7.

25. 5 bells are heard at intervals of 3, 6, 8, 10 and 12 minutes respectively. If all the bells toll together exactly at 2:23 PM, they will again be heard together at ? **(Cogent)**

A) 1:50 pm **B)4:23pm** C)1:20 pm D) 2:50 pm

Sol: Application of LCM
LCM of (3, 6, 8, 10, 12)= 120
So after 120 minutes(i.e., 2 hours) from 2.23pm again they will toll together = 2:23 pm + 2 hrs= 4:23pm

Practice:

1. Find the largest 3 digit number which leaves a remainder of 6 when divided by 24, 30 and 40.
   a) 978 b) 966 c) 996 d) 963
2. Find the smallest number which leaves a remainder of 3 when divided by 12 and leaves a remainder of 6 when divided by 15.
   a) 63 b) 27 c) 33 d) 51
3. If the H.C.F of two numbers is 15 and their products is 16200 find the L.C.M of the numbers.
   a) 162 b) 108 c) 720 d) 1080
4. Find the smallest number which leaves a remainder of 3 when divided by 5 and leaves a remainder of 7 when divided by 9.
   a) 47 b) 43 c) 45 d) 34
5. Find the greatest 3 digit number which leaves a remainder of 3 when divided by 15 and leaves a remainder of 12 when divided by 24.
   a) 972 b) 993 c) 948 d) 978
6. The L.C.M of $2^3 \times 3^2 \times 5$ and $3^3 \times 5^2 \times 2^3$ is
   a) $2^2 \times 3^2 \times 5$ b) $2^3 \times 3^3 \times 5^2$ c) $2^3 \times 5 \times 3^2$ d) $2^3 \times 3^2 \times 5$
7. Find the least number which leaves respective remainders of 2, 3 and 4 when divided by 8, 9 and 10.
   a) 354 b) 366 c) 362 d) 358
8. Find the smallest five-digit number which leaves respective remainders of 3, 9, 15 and 27 when divided by 24, 30, 36 and 48.
   a) 10101 b) 10720 c) 10789 d) 10059
9. Find the greatest two digit number which leaves respective remainders of 3 and 7 when divided by 12 and 16 respectively.
   a) 85 b) 86 c) 87 d) 89
10. If the H.C.F of two numbers is 8 and the product of the numbers is 768, how many such pairs are possible?
    a) 1 b) 0 c) 2 d) 3
11. What is the greatest number of 3 digits which when divided by 6, 9 and 12 leaves a remainder of 3 in each case?
    a) 903 b) 939 c) 975 d) 996
12. The largest four-digit number which when divided by 4, 7 or 13 leaves a remainder of 3 in each case, is
    a) 8739 b) 9831 c) 9834 d) 9893
13. Let the least number of six digits, which when divided by 4, 6, 10 and 15, leaves in each case the same remainder of 2, be N. The sum of the digits in N is
    a) 3 b) 4 c) 5 d) 6

14. Find the greatest number which when it divides 367, 135 and 259 will leave the same remainder in each case.
    a) 2 b)4 c)6 d)8
15. Find the least number which when divided by 5, 7 and 11, leaves a remainder 6 in each case.
    a) 379 b)381 c)391 d)400
16. Find the largest number which can divide 4059, 4264 and 4182.
    a) 79 b)41 c)91 d)40

17. Three numbers are in the ratio of 3: 4: 5 and their L.C.M. is 2400. Their H.C.F. is **(TCS)**
    a) 40 b) 80 c) 120 d) 200
18. The H.C.F. and L.C.M. of two numbers are 21 and 4641 respectively. If one of the numbers lies between 200 and 300, the two numbers are **(Amazon)**
    a) 273, 357 b) 273, 359 c) 273, 361 d) 273, 363
19. 21 mango trees, 42 apple trees and 56 orange trees have to be planted in rows such that each row contains the same number of trees of one variety only. Minimum number of rows in which the trees may be planted is **(Tech Mahindra)**
    a) 3 b) 15 c) 17 d) 20
20. Three sets of English, Mathematics and Science books containing 336, 240 and 96 books respectively have to be stacked in such a way that all the books are stored subject wise and the height of each stack is the same. Total number of stacks will be **(Capgemini)**
    a) 14 b) 21 c) 22 d) 48
21. Four metal rods of lengths 78 cm, 104 cm, 117 cm and 169 cm are to be cut into parts of equal length. Each part must be as long as possible. What is the maximum number of pieces that can be cut? **(HCL)**
    a) 27 b) 36 c) 43 d) 480
22. The number between 4000 and 5000 which is divisible by 12, 18, 21 and 32 is **(CTS)**
    a) 4023 b) 4032 c) 4203 d) 4302
23. The number nearest to 43582 divisible by each of 25, 50 and 75 is **(CTS)**
    a) 43500 b) 43550 c) 43600 d) 43650
24. Find the least number which when divided by 12, leaves a remainder of 7; when divided by 15, leaves a remainder of 10 and when divided by 16, leaves a remainder of 11. **(CTS)**
    a) 115 b) 235 c) 247 d) 475
25. The greatest number which when subtracted from 5834, gives a number exactly divisible by each of 20, 28, 32 and 35 is **(Amazon)**
    a) 1120 b) 4714 c) 5200 d) 5600
26. The least number which is a perfect square and is divisible by each of the numbers 16, 20 and 24, is **(TCS)**
    a) 1600 b) 3600 c) 6400 d) 14400

27. The smallest number, which when diminished by 7, is divisible by 12, 16, 18, 21 and 28 is **(CISCO)**

a) 1008 b) 1015 c) 1022 d) 1032

28. The least number which when increased by 5 is divisible by each one of 24, 32, 36 and 54 is **(CTS)**

a) 427 b) 859 c) 869 d) 4320

29. The least number, which when divided by 12, 15, 20 and 54 leaves in each case a remainder of 8 is **(Capgemini)**

a) 504 b) 536 c) 544 d) 548

30. A number less than 500, when divided by 4, 5, 6, 7 leaves remainder 1 in each case. The number is **(Wipro)**

a) 211 b) 420 c) 421 d) 441

KEY

1. B	2. D	3. D	4. B	5. C
6. B	7. A	8. A	9. C	10. C
11. B	12. B	13. C	14. B	15. C
16. B	17. A	18. A	19. C	20. A
21. B	22. B	23. D	24. B	25. B
26. D	27. B	28. B	29. D	30. C